Flow Sensing & Power Generation for IoT

Watts Water Technologies

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1. Overview & Background

This document is a master reference for flow sensing and power generation research conducted at **Watts Water Technologies**.

It covers flow metering technology selection, micro hydro turbine generator analysis, a classification system for turbine types, testing methodology, load strategies, and a catalogue of all devices evaluated. A turbine selection framework and worked application examples are provided at the end.

Design priorities across all applications:

- Low cost
- Accurate at low flow rates
- Low power consumption
- Potential for self-powered IoT (no wired power, no battery replacement)

1.1 Why Rotary? Why 0-4 GPM?

The scope of this document is **0-4 GPM**. This range covers the fixtures of interest: bottle fillers, drinking stations, kitchen sinks, restroom sinks, and residential showers. Most single fixtures fall below 2 GPM; the 4 GPM ceiling accommodates multi-fixture configurations such as up to four sinks sharing one sensor, without changing the technology selection.

Rotary sensing measures volume directly by counting spinner revolutions rather than inferring flow from pressure. At low flow rates, ΔP across any restriction is too small to measure reliably; a spinning turbine at 0.5 GPM produces a clear, countable pulse train. Above 4 GPM, differential pressure sensing takes over as the practical choice.

1.2 Prior Work

An initial IoT Technology Readiness Report was prepared at Bradley, which benchmarked and tore down competitor faucets (U by Moen, Delta Essa VoiceIQ), independently confirming the GEMS 238600 turbine in both and validating it as the optimal sensing element. That report also identified differential pressure sensing as a viable approach for higher-flow products (EFX emergency showers at 15-30 GPM), and suggested continued research into turbine generators for self-powered applications, which is where this document continues.

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2. Flow Sensing

2.1 Technology Survey

Technology	Principle	Strengths	Weaknesses	Best Fit
Rotary (Turbine / Paddle)	Spinning element; RPM proportional to flow	Low cost, good low-flow accuracy, insertable	Moving parts, wear	0.08-8 GPM
Positive Displacement	Fluid fills/empties fixed chambers	Good at low flow	Many moving parts, high ΔP, complex housing	0.25-2 GPM
Differential Pressure	ΔP across restriction proportional to Q^2	No moving parts, cheap sensors	Poor at low flow/pressure	>4 GPM
Ultrasonic	Transit time of sound	Non-invasive, no ΔP	Expensive, special housing	High accuracy needs
Magnetic	Faraday's law on conductive fluid	Non-invasive, accurate	Needs conductive particles, expensive	Industrial

Technology	Principle	Strengths	Weaknesses	Best Fit
Thermal / Coriolis	Heat transfer or mass flow	Very accurate	Expensive, power-hungry	Not in scope
Vortex Shedding	Oscillating vortices behind bluff body	Accurate, no moving parts	Min flow >2 GPM, high ΔP, long length	>2 GPM
Vane / Piston	Spring-loaded vane deflects with flow	Wide range, low-flow capable	Mechanical complexity, spring	Similar to turbine

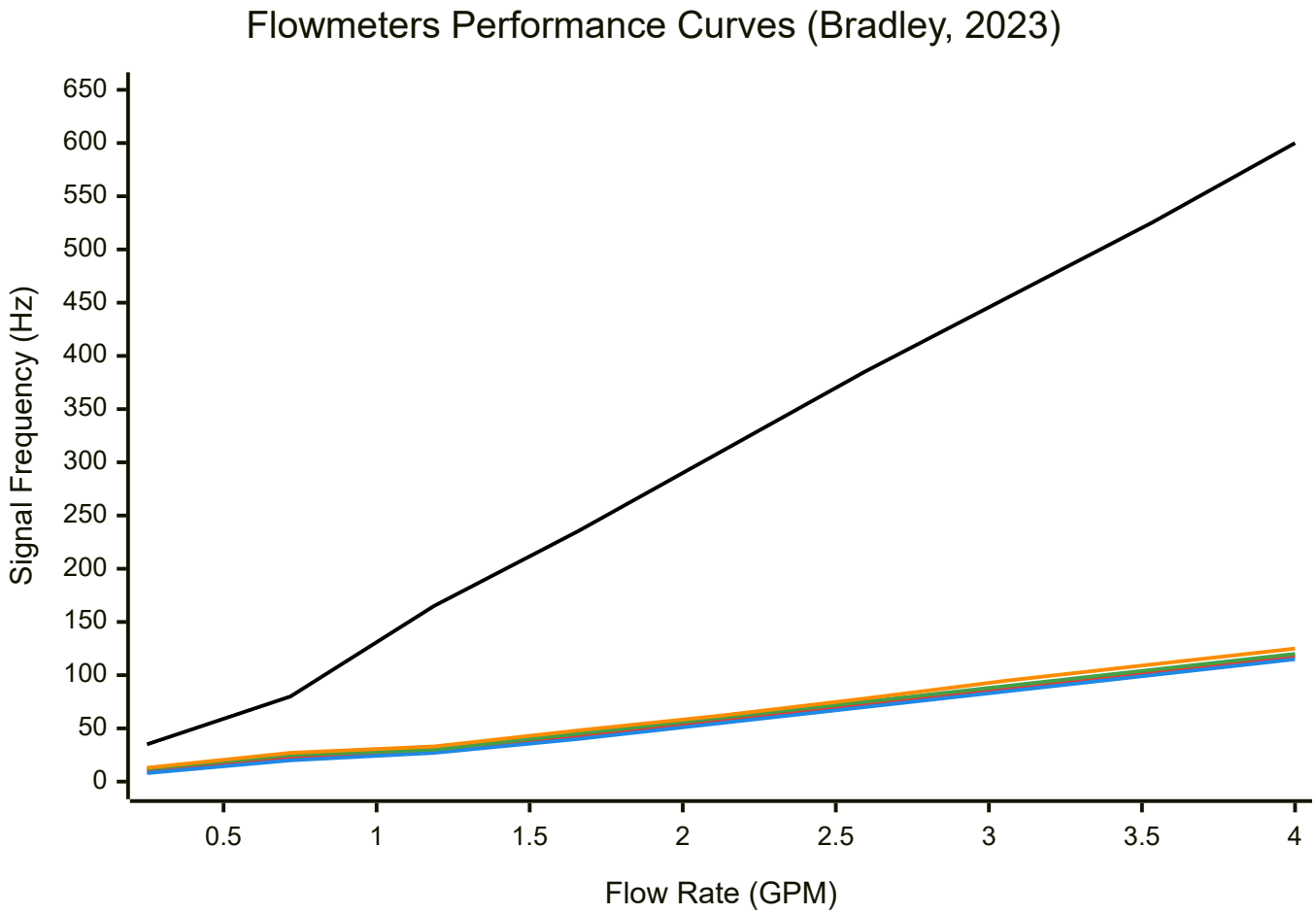
2.2 Rotary Sensors: Radial vs. Axial

Type	Description	Min Flow	Notes
Radial / Paddle Wheel	Only part of rotor in flow path	~0.1-0.25 GPM	Cheaper, less accurate at low flow
Axial / Inline	Full propeller inline; Hall effect reads blades	~0.05-0.08 GPM	Better low-flow sensitivity

Paddle-wheel meters (Gredia, Sea YF-S201, Uxcell) require higher flow to start spinning. Axial turbines (GEMS 238600, Sika VY10, Auk Mueller MT5) offer superior low-flow performance. The GEMS 238600, available as a turbine element or full assembly, was selected as the primary sensing element based on in-house testing and competitor validation.

Flowmeters Performance Curves

Signal frequency vs. flow rate for five candidate rotary flow meters. All relationships are linear ($f = k \cdot Q$), as expected from direct pulse counting. The **GEMS 238600** produces roughly 4-5× the frequency-per-GPM of any other meter in the set, which is the primary reason it was selected: more pulses per unit volume means higher resolution at low flow, which is where fixture measurement lives.



Legend:

	Meter	Slope (Hz / GPM)
	GEMS 238600	~150
	Sea YF-S201	~30
	Gredia FS200A	~29
	Sika VY10	~31
	FS200A	~32

Data reproduced from Bradley in-house testing, 2023.

2.3 Hall Effect Sensors

The turbine's embedded magnet is read by an external Hall effect sensor. The GEMS 238600 emits approximately **100 Gauss** at the sensor zone (measured in-house against a reference Hall sensor).

Sensor	Supply Voltage	Active Current	Active Power	Sleep Current	B_{OP} (Operate)	B_{RP} (Release)	Key Advantage
Allegro A1220	3-24V	4 mA (always on)	~12 mW @ 3V	N/A	35 G typ	15 G typ	Proven, wide voltage range
Allegro A1171	1.65-3.5V	3.5-12 µA	~15-42 µW	8 µA	45 G typ	30 G typ	~800x lower power than A1220

Both operate comfortably with the 100 G field from the GEMS turbine. The A1171 is preferred for any battery-powered or energy-harvesting application.

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3. Turbine Generators

This section introduces turbine generators and classifies them by physical form. The design-constraint framework and all experimental characterization follow in §4.

3.1 Goals & Operating Range

Many under-sink or inline applications need IoT capability but have no wired power. The goal is to generate usable electrical power from water flow at **0.5-1.0 GPM**, well below where most off-the-shelf generators are optimized.

Generators can optionally double as flow sensors since their output frequency and power output are both proportional to flow rate. This is covered in §4.5.

3.2 Magnet Magnetization

All rotating magnets in these generators must be **diametrically magnetized**: N/S poles on opposite sides of the magnet, perpendicular to the axis of rotation. As the magnet spins, the field alternates between N and S, inducing voltage. An axially magnetized magnet spinning on its own axis presents a constant field to a symmetric coil, producing no change in flux and no induced voltage. This is not a design variable; all generators in this catalogue use diametric magnetization.

3.3 Admission Types

Both full and partial admission use a directed jet of water to spin the rotor, but differ significantly in geometry and performance:

Type	Description	Key Characteristic
Full Admission	Tight-tolerance nozzle directs a jet around the entire rotor circumference; all blades in flow at all times	Lower startup flow, higher ΔP, precise machining required
Partial Admission	Simpler, smaller inlet hole; jet hits only a portion of the rotor at any moment	Higher startup flow, lower ΔP, simpler geometry

Full admission generators can start generating at lower flow rates, critical for the 0.5 GPM target. Partial admission hub generators sit outside the direct flow path and require a cage or nozzle structure to redirect the jet onto the blades.

3.4 Coil Configurations & Flux Types

Flux Types

	Axial Flux	Radial Flux
Flux vs rotation axis	Parallel (flux lines run along the axis of rotation)	Perpendicular (flux lines run across the axis of rotation)
Typical coil shape	Pancake (claw pole) or wrapped cylinder (belt)	Small rings pointing inward, or ring-in-ring
Flux guides needed?	Yes, always required for both claw pole and belt variants	Typically no (spoke/gyro)
Cycles per revolution	Equal to the number of magnet pole pairs, matched by the number of claw-pole pairs in the flux guide	Equal to the number of magnet pole pairs

Coil Configurations

Our Name	Literature Name	Flux Type	Description	Example
Claw pole	Claw pole	Axial	Pancake coil above a cage of soft-iron claw legs that funnel flux from the spinning magnet through the coil. The number of claw poles sets the AC frequency per revolution.	Zurn P6900
Belt	Axial flux wrapped	Axial	Coil wraps around the rotor cylinder, enclosing the magnet.	Toto EcoPower insert, Toto 10s Dynamo (EDV462/EDV561)

Our Name	Literature Name	Flux Type	Description	Example
			Best flux coupling of the axial types.	
Spoke	Hub generator	Radial	Multiple small coils facing inward; magnet ring spins outside them (outrunner).	F50
Gyro	Radial flux ring	Radial	Ring spinning inside another ring. Streamlined for in-pipe mounting.	M6 axial propeller

Belt Sub-Categories

- **Single coil:** One electrical circuit, two wires. e.g., Toto EcoPower insert.
- **Dual coil:** Two independent circuits, four wires. One coil is dedicated to always-on critical functions (solenoid valve, MCU); the other powers secondary functions (display, radio) that can be shed when insufficient power is being generated. The Toto 10s Dynamo (EDV462/EDV561) uses this design with a high-inductance secondary coil. Toto advertises that just 10 seconds of water flow generates enough stored energy to fully initialize the electronics, with the inductance sustaining power briefly after flow stops to ensure clean valve closure and final sensor reads. The tradeoff is that high inductance increases braking on the rotor.

Three-Phase (Spoke/Hub)

Six diodes in hub generators confirm 3-phase generation (3-phase bridge rectifier). Benefits over single phase: smoother and more continuous power delivery, even rotor braking. Reduced DC ripple is useful for stable power delivery but a disadvantage if ripple frequency is being used to infer rotor RPM. In that case a single-phase or unfiltered output is preferable (see §4.5).

> **Note on coupling:** The real-world flux linkage between magnet and coil depends on geometry, air gap, material permeability, and other factors that are difficult to measure directly without a gaussmeter. For comparative analysis between turbines, this is treated as a lumped experimental parameter derived from open-circuit testing rather than calculated from first principles.

3.5 Physics: Power Generation

What Drives Output Power

At a fixed load resistance, the voltage induced in the coil depends on coil turns (N), magnet strength (B), coil area (A), and rotor angular velocity (ω). In the **linear regime** (moderate ω, no

magnetic saturation, rotor not yet braking-limited) output power scales with V^2 and therefore approximately with ω^2 and Q^2. At higher flow rates this breaks down: voltage flattens as electromagnetic braking, bearing drag, and rotor hydrodynamic losses grow faster than available torque, so power rises sub-quadratically and eventually asymptotes.

Increasing any of N, B, A, ω increases both output voltage and electromagnetic braking on the rotor. Of those four variables, each increases output and braking together. Two separate levers help without a direct braking penalty:

- **Thicker wire (lower R_{coil}):** Lowers coil resistance without changing magnetic coupling. Less power dissipated internally as heat; more delivered to the load.
- **More water pressure:** Increases torque available to overcome braking. Pure energy input.

Why Each Generator Has a Power Sweet Spot

At very low flow rates, friction and bearing losses consume a disproportionate share of the available hydraulic power, and electrical output is low. As flow increases, output power rises. Electromagnetic braking also rises with rotor speed, and at high flow rates the braking torque becomes significant enough that additional water energy is increasingly lost rather than converted. Efficiency therefore peaks at an intermediate flow rate, the turbine's natural sweet spot.

This sweet spot is turbine-specific and determined experimentally. Some turbines, particularly the axial propeller type, show no interior sweet spot: efficiency decreases monotonically from the lowest tested flow rate, meaning the lowest flow rate is always the most efficient operating point. Both cases are handled by the load-strategy framework in §4.3.

Pressure vs. Flow Rate

$$P_{hydraulic} = \Delta P \cdot Q$$

Pressure (ΔP) provides torque; flow rate (Q) provides rotor speed. Both are needed. A high-pressure, zero-flow system produces no power, and a high-flow, zero-pressure system produces no power. At a fixed load, output power rises with both but is ultimately bounded by the rotor's mechanical limits at high flow rates.

Efficiency

$$\eta = \frac{P_{electrical}}{P_{hydraulic}} = \frac{P_{load}}{\Delta P \cdot Q}$$

> Convert GPM to m³/s and PSI to Pa for SI unit consistency.

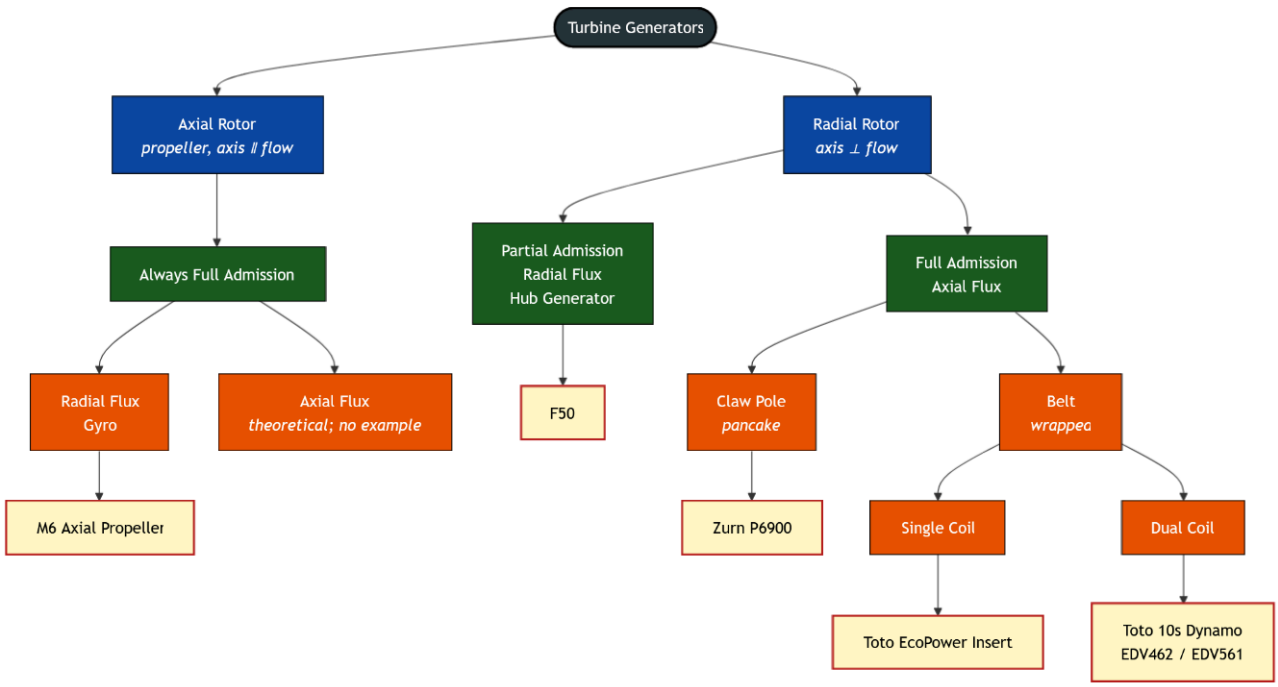
3.6 Classification System & Family Tree

Turbine generators are classified by rotor orientation, admission type, flux direction, and coil configuration. **These axes are independent.** An axial rotor could theoretically pair with axial flux coils, though no commercial example in this scope has been found.

Rotor Types

- **Axial rotor (propeller):** Axis parallel to water flow. Water hits all blades simultaneously. Always full admission.
- **Radial rotor:** Axis perpendicular to water flow. Can be full or partial admission.

Family Tree



Legend: rotor type · admission · flux / coil configuration · commercial example

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4. Design Constraints & Generator Modification

Classification tells us what candidate devices exist. This section establishes the criteria that govern selection (§4.1), describes the bench procedure that populates those criteria (§4.2), turns the resulting data into a fixed load resistance per turbine (§4.3), walks the full procedure on the Zurn P6900 as a worked example (§4.4), and covers the generator-output types and modifications required to use the generator as a flow sensor (§4.5-§4.7).

4.1 Completing Criteria for Rotary Flowsense Selection

To select a device for a given product, two time-varying profiles and a set of constraint inputs must be matched against the device's measured capabilities.

Governing profiles:

- **Power demand over time.** How much power does the product need, and when? Always-on, only during flow, or bursty (radio transmit)?
- **Flow over time.** How much water passes through, and when? Continuous, intermittent, or dormant for long periods?

Constraint inputs: pressure budget, price budget, sensing accuracy, physical envelope, and any application-specific requirements (NSF, PID compliance, fitting type, sealed housing).

Shortcut for simple cases. If the product allows a wired supply or a replaceable battery, the answer is a GEMS 238600 paired with an Allegro A1220 (wired) or A1171 (battery) Hall sensor. The framework below applies to the harder case of sealed, long-life, or self-powered deployments, though it handles the simple cases as well.

Five Output Configurations

Every rotary flowsense decision resolves to one of the following:

#	Configuration	Sensing	Power	Handles Zero Flow	Handles Ultra-Low Flow	Appendix Fields Queried
1	Flow meter + battery	Dedicated meter (GEMS + A1171 or A1220)	Battery (or wired)	Yes (sleep)	**Ultra-low** (GEMS min ≈ 0.08 GPM)	App. B: min flow, active/sleep power
2	Turbine alone	None	Turbine output	No	**None** (regulated turbine)	App. C: min flow, R_{eff} or R_{max}, peak power
3	Turbine as sensor	Power at fixed load, or AC / ripple frequency	Turbine output	No	**Medium** (turbine as sensor, limited by loaded Q_{start})	App. C: R_{eff}, peak power, ΔP profile
4	Turbine + internal Hall	Internal Hall pulses (load-independent)	Turbine output	Yes (battery)	**Low*** (turbine with Hall,	App. C: min flow, Q_{start}, peak power

#	Configuration	Sensing	Power	Handles Zero Flow	Handles Ultra-Low Flow	Appendix Fields Queried
			+ small battery		open-circuit sensing; additional testing required per turbine)	
5	Paired meter + turbine + battery	Dedicated meter	Turbine + battery trickle-charge	Yes (battery)	**Ultra-low** (meter sets the floor)	App. B + App. C

**Configuration 4 pushes minimum detectable flow below the turbine's loaded Q_{start}, but exactly how low depends on bearing drag, magnetic detent, and rotor inertia at the Hall sensor's threshold. This must be measured per turbine as part of the bench procedure in §4.2.*

Two Cross-Cutting Robustness Requirements

Every configuration must be evaluated against two failure modes:

- **Zero-flow events.** No water moving. The system must remain responsive (wake-on-flow) without draining the battery or losing state. Addressed by sleep mode (§5.3) and a low-power wake comparator.
- **Ultra-low-flow / insufficient-power events.** Flow is present but below the turbine's usable threshold. The generator produces voltage but not enough to run the load. Addressed by open-circuit sensing mode (§4.7) and stored-energy buffering (§5.2).

Configurations 1, 4, and 5 handle both modes. Configurations 2 and 3 fail silently during either mode and are only appropriate where continuous minimum flow is guaranteed.

4.2 Testing & Experimental Procedure

The procedure below is run once per turbine and produces the raw data that populates Appendix C, feeds the load-strategy selection in §4.3, and ultimately enables the selection algorithm in §6.

Test Rig Setup

1. Set target flow rate (Q)
2. Measure ΔP across the device

3. With the coil leads open (no load), sweep flow up from zero and record the minimum Q at which any non-zero voltage appears across the open circuit. The voltage magnitude is not important, only that it is readable; a non-zero reading confirms the rotor has started spinning. Also record ΔP vs. Q with no load as the hydraulic-only baseline.
4. Measure R_{coil} (done once per turbine)
5. Connect variable resistor; sweep full range at each Q, recording V_{load}, current, frequency
6. Calculate efficiency at each (Q, R) point
7. Step up flow rate and repeat from 0-4 GPM

> **Note on regulated-DC units:** Turbines with built-in regulators (e.g., Zurn P6900) are tested by opening the housing, disconnecting the internal battery and regulator, and tapping the two raw coil leads directly. The regulator flattens flow-dependent information and cannot be characterized by an external R-sweep. For off-the-shelf use the built-in regulator is the product; for benchtop characterization the raw coil output is what gets swept.

Measurements Per Flow Rate

Measurement	Symbol	Notes
Flow rate	Q (GPM)	Set value
Differential pressure (loaded)	ΔP	Across the device under a connected load
Differential pressure (open circuit)	ΔP_{oc}	Across the device with coil leads open; hydraulic-only baseline
Minimum startup flow (open circuit)	Q_{start}	Lowest Q at which any non-zero voltage appears across the open circuit
Coil resistance	R_{coil}	DC, measured once per turbine
Load resistance (swept)	R_{load}	Full range at each Q
Voltage at each R_{load}	V_{load}	Measured across the load
Power at each R_{load}	P	Calculated: V^2/R_{load}
Efficiency at each R_{load}	η	Calculated: $(V^2/R_{load})/(\Delta P \cdot Q)$
AC frequency	f	Proportional to flow rate; usable for sensing (see §4.5)

Why Open-Circuit Testing Matters

The open-circuit sweep establishes the absolute minimum flow at which the rotor will spin with zero electromagnetic braking. This is the floor for open-circuit sensing mode (§4.7), where the

load is disconnected and the system runs on a small battery to read rotor pulses at very low flow. Whatever minimum flow the turbine achieves under load, it will achieve a lower minimum with the load removed. That delta is the reason open-circuit sensing is worth the extra MOSFET and battery.

Experimental Isolation Procedure

Step	What	Method	Variables Isolated
1. Teardown	Direct measurement	Count turns/poles, measure with multimeter/calipers	N, A, r, R_{coil}, coil geometry
2. Open circuit	No load	Measure frequency and ΔP_{oc} at several Q points	lumped flux coupling, friction/efficiency ratios
3. Loaded	Known R_{load} connected	Measure V_{load}, I, ω; back-calculate	$\eta_{turbine}$
4. Post-processing	Separate ratios using $\eta_{turbine}$	Compute from step 2 and step 3 data; subtract known losses to isolate remaining friction terms	c_f (viscous friction), τ_0 (static friction)

Can measure directly: R_{coil}, wire gauge, pole/claw count, coil ID/OD, rotor type, admission type, flux type.

Cannot currently measure: B (no gaussmeter), L (LH meter TBD). Since B cannot be measured directly, the flux coupling is treated as a single lumped experimental constant.

Interpolating Between Tested Resistances (Optional)

All reported R_{eff}, Q_{eff}, R_{max} values and all Cost/Reward charts use measured $(R, V, Q, \Delta P)$ points directly with no extrapolation. In principle, a Thevenin source model

$$V_{load} = V_{source} \cdot \frac{R_{load}}{R_{source} + R_{load}}$$

fit to a few measured points per flow rate could predict V at untested resistances. In practice, finding R_{eff} requires a dense R-sweep at multiple Q anyway to locate the efficiency peak, so the model provides no testing shortcut. It would only be useful for generating smooth visual curves after the fact. Measured points are sufficient for all selection decisions.

4.3 Load Strategies

Once the bench data is in, the next question is what resistor to connect to the coil in the product. This is an experimental decision that cannot be predicted from geometry alone because it depends on coil impedance, rotor drag, and magnetic coupling that cannot all be

measured directly (see §4.2). The R-sweep at each flow rate produces a family of efficiency and power curves, and the load-strategy framework below converts those curves into one fixed resistor value per turbine.

Picking the flow rate that produces the most raw power is tempting but wrong. Raw power keeps climbing well past the point where ΔP has blown through any realistic pressure budget, and voltage flattens as the rotor approaches its electromagnetic limit, meaning additional water energy yields diminishing electrical returns. The useful operating point is the one that balances power out against hydraulic cost: the efficiency peak.

Key Terms

Symbol	Name	Definition
R_{eff}	Peak-efficiency resistance	Single fixed resistance at which the turbine achieves greatest efficiency across all tested flow rates. One value per turbine.
Q_{eff}	Peak-efficiency flow rate	Flow rate at which that greatest efficiency occurs. The turbine's natural sweet spot.
R_{max}	Asymptotic optimal resistance	Applies only when minimum tested flow rate is also the most efficient (no interior sweet spot). The asymptote approached by optimal R as Q increases. Used as the fixed load for these turbines.

Selection Procedure

Step 1. Open-circuit baseline. With the coil leads open, sweep flow up from zero and record (a) Q_{start}, the minimum flow at which any non-zero voltage appears across the open circuit, and (b) ΔP_{oc} vs. Q, the hydraulic-only baseline with no electromagnetic braking.

Step 2. Measure internal resistance. Measure R_{coil} with a multimeter. The experimentally optimal load will typically be higher than R_{coil} because the coil's AC source impedance includes inductive reactance (ωL) in addition to R_{coil}. The true optimum is determined by the sweep in Step 3, not by a simple DC impedance-matching rule.

Step 3. Efficiency vs. resistance sweep. At each flow rate Q, sweep load resistance and calculate:

$$\eta = \frac{V_{load}^2 / R_{load}}{\Delta P \cdot Q}$$

This produces a family of curves, one per flow rate, plotted on the same axes.

Step 4. Identify R_{eff} and Q_{eff} (or R_{max}). Find the single point of greatest efficiency across all (Q, R) combinations.

- **Interior sweet spot found:** that resistance and flow rate are R_{eff} and Q_{eff}. Use R_{eff} as the fixed load.
- **Minimum flow rate is always most efficient:** no interior sweet spot. Plot optimal R vs. Q; the curve approaches an asymptote, R_{max}. Use R_{max} as the fixed load. Selection is then driven by pressure budget or power need.

Sanity check. The voltage vs. flow curve should begin to flatten at or near Q_{eff}. This is the same electromagnetic saturation behavior that makes operation at higher flow rates wasteful, and serves as an independent confirmation of the efficiency-based peak.

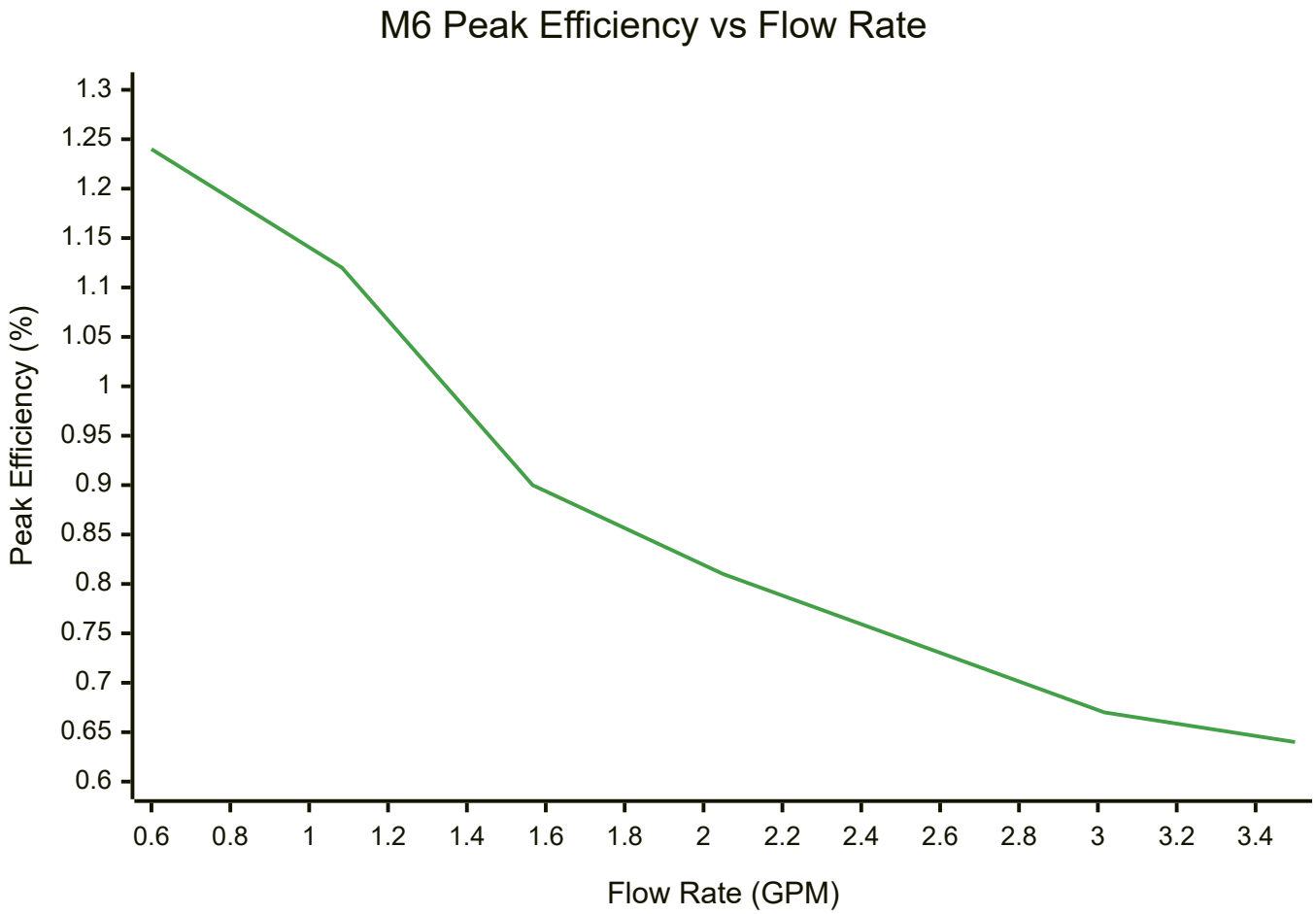
Flatness observation. Changing load resistance within a reasonable range around the optimum typically produces only modest changes in power and efficiency. A well-chosen fixed load captures nearly all the available benefit without the cost and complexity of a real-time dynamic system, though this should be verified per turbine as data becomes available.

Fixed Load Selection Rule

Turbine Type	Fixed Load	Notes
Has interior Q_{eff}	R_{eff}	Best efficiency at the turbine's natural sweet spot
Min flow = Q_{eff}	R_{max}	No single optimal flow rate; selection by pressure budget or power need

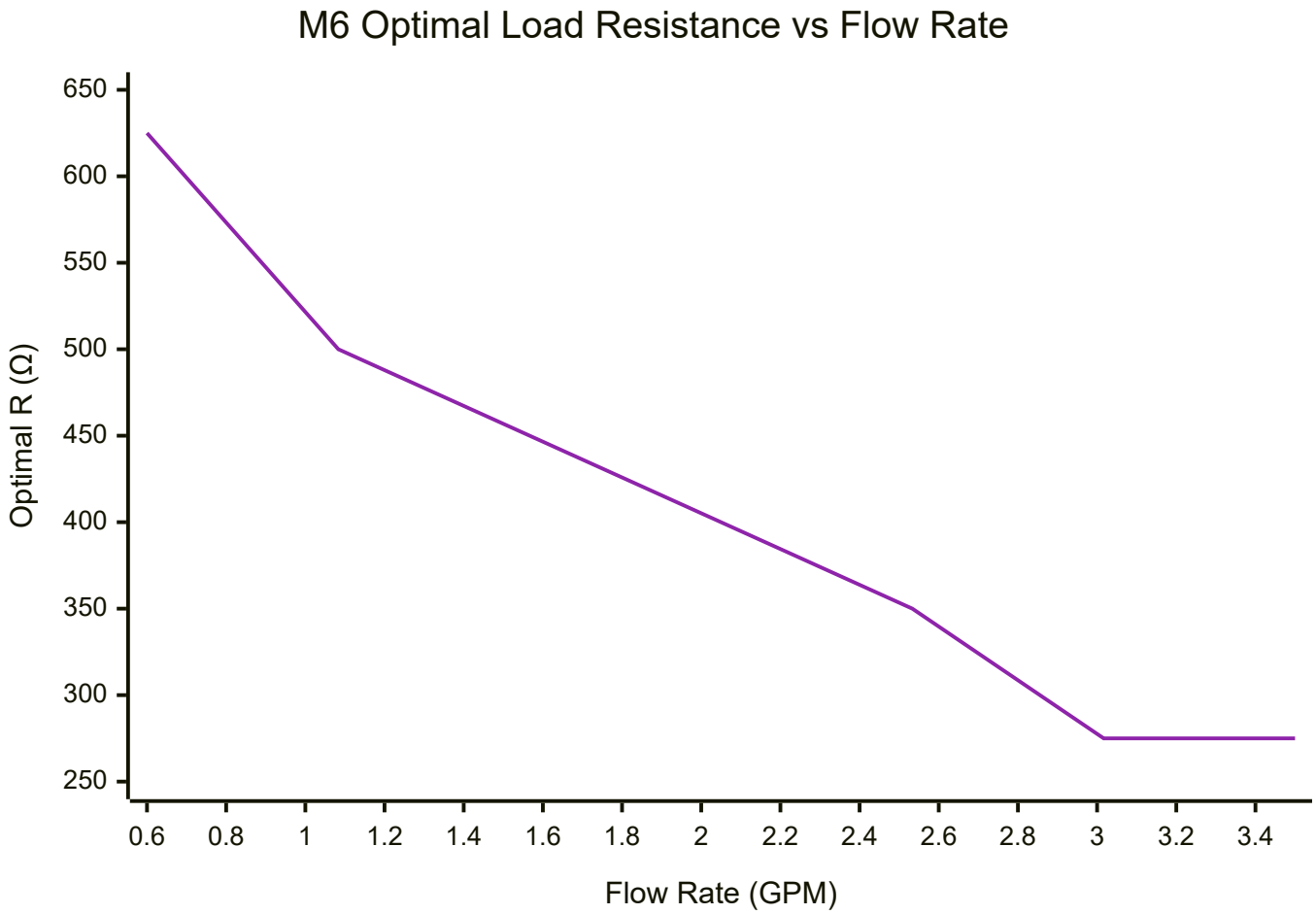
Worked Example: M6 Axial Propeller (the R_{max} Case)

The M6 axial propeller illustrates the second branch of the rule. Full R-sweep at seven flow rates (0.60-3.50 GPM, $R_{coil} = 161.6\ \Omega$) shows **no interior efficiency peak**. Efficiency is highest at the lowest tested flow rate and decreases monotonically as Q rises, which is characteristic of axial-propeller geometry.



Peak efficiency at each tested flow rate; $R_{coil} = 161.6\ \Omega$.

Because no single flow rate is best, the load is chosen by looking at the optimal resistance at each Q. Optimal R decreases with increasing Q and approaches an asymptote near $275\ \Omega$.



Optimal load resistance at each tested flow rate; asymptote at $R_{max} \approx 275\ \Omega$.

$R_{max} \approx 275\ \Omega$ is the fixed load selected for the M6. Because peak efficiency occurs at the low-flow end of the range, final operating-point selection is driven by the pressure budget and power need for the target application rather than by a single sweet-spot flow rate. M6 peak power ranges from ~3 mW at 0.60 GPM ($\Delta P \approx 1$ PSI) to ~490 mW at 3.50 GPM ($\Delta P \approx 50$ PSI), with efficiency steadily degrading across that range.

Contrast with Zurn P6900 (§4.4): the Zurn has a clear interior Q_{eff} at 0.33 GPM and uses a single R_{eff}. The M6 has no interior peak and uses an R_{max} asymptote. These two turbines demonstrate both branches of the selection rule in practice.

Pre-Programmed Dynamic Load (Optional)

A lookup table indexed by flow rate can switch between a small set of resistors, populated from bench test data and not real-time. Adds ~ $2-5BOM. Onlyworthconsideringwhentheoperatingrangespans > 3:1flowratioandoptimal\$R$ shifts dramatically across that range. For most 0-4 GPM applications, a fixed R_{eff} or R_{max} is preferred.

Standard Output Charts Per Turbine

Once the fixed load is chosen, each turbine is characterized by two summary quantities plotted together on a single **normalized Cost/Reward chart**:

- **Cost:** ΔP vs. Q, the hydraulic cost imposed on the system
- **Reward:** Power vs. Q, the electrical output at the chosen fixed load

Both are expressed as percent of their respective turbine-specific maximum, so they share a single axis. Conversion factors back to engineering units are given per-turbine in the caption. §4.4 shows the full workflow applied to the Zurn P6900.

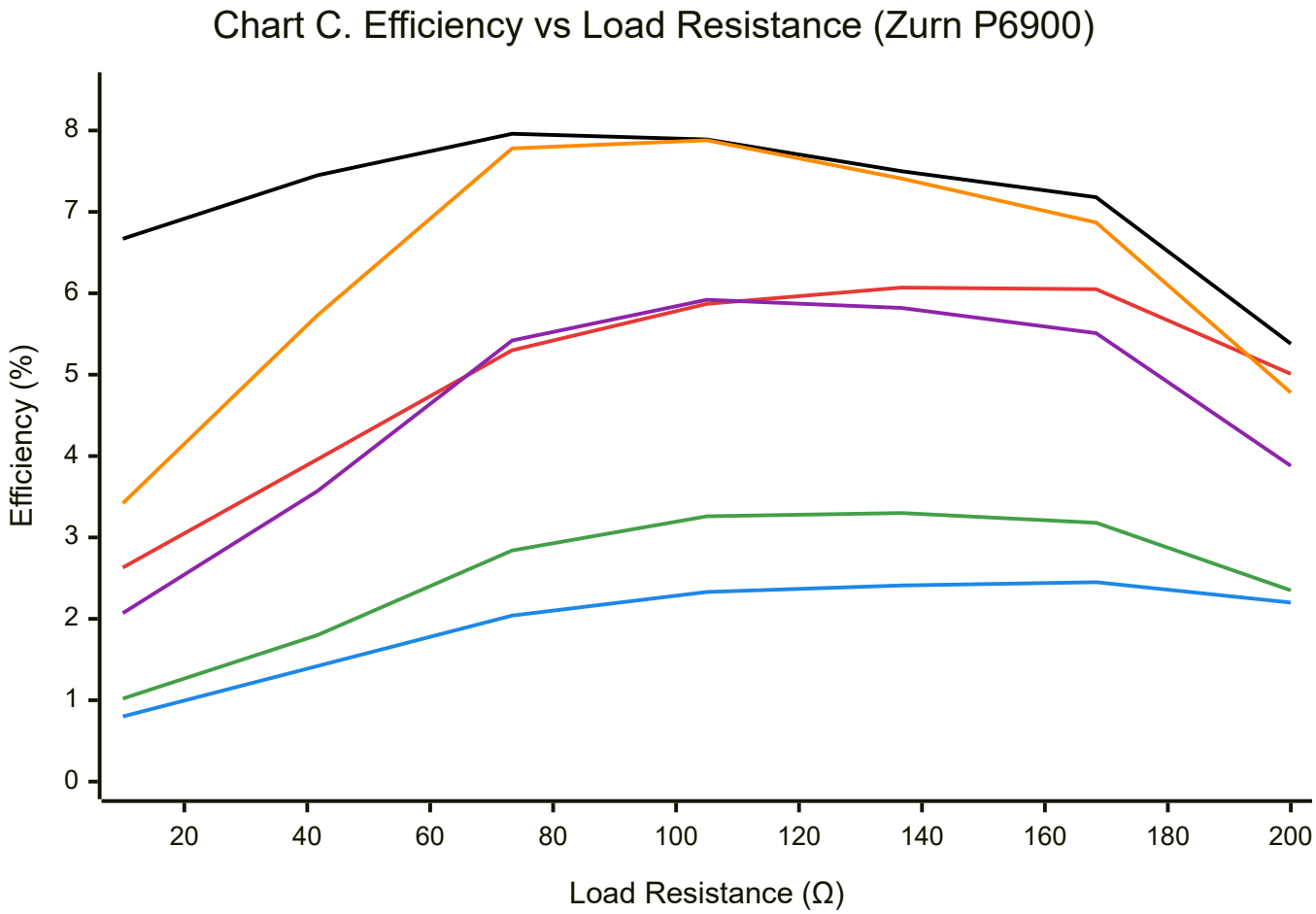
4.4 Test Results: Zurn P6900 Demo

The Zurn P6900 is the primary demonstration unit with complete bench data including full flow rate sweep and full resistance sweep at each flow rate. The unit is natively regulated DC; these results were taken with the battery and regulator disconnected, tapping the two raw coil leads directly (see §4.2 note). It serves as a concept demonstrator (good operating range, clean ΔP vs Q curvature, clear interior Q_{eff}), not a product recommendation. This is the R_{eff} / Q_{eff} branch of the selection rule; for the contrasting R_{max} case, refer to the M6 worked example in §4.3. All other turbines in Appendix C will follow this same format as data becomes available.

Measured coil resistance: $R_{coil} = 3.6\ \Omega$.

Step 1: Efficiency vs. Load Resistance (Finding R_{eff} and Q_{eff})

The efficiency sweep produces one curve per flow rate. The global peak across all (Q, R) combinations identifies R_{eff} and Q_{eff}.



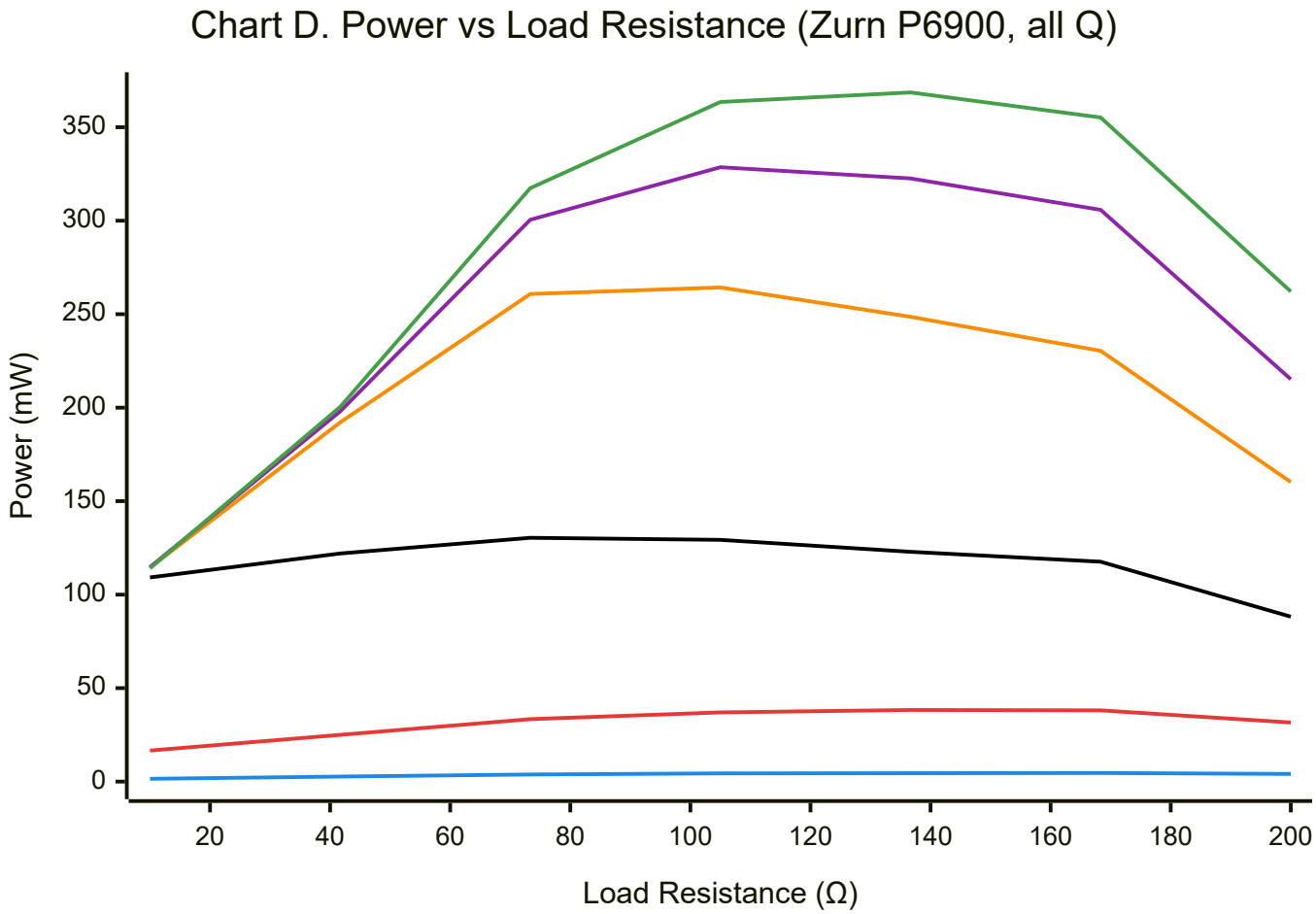
Legend:

	Flow Rate	Peak η
	0.17 GPM	2.5%
	0.25 GPM	6.1%
	0.33 GPM (Q_{eff})	**8.0%**
	0.42 GPM	7.9%
	0.50 GPM	5.9%
	0.67 GPM	3.3%

The global maximum sits on the $Q = 0.33$ GPM curve at $R = 40\ \Omega$, with $\eta \approx 8.0\%$. No higher efficiency is reached at any other flow rate or resistance in the tested range.

Step 2: Power vs. Load Resistance (Confirming Flatness)

Plotting power at the same (Q, R) grid confirms that the optimum is broad. Small deviations from R_{eff} produce only modest power changes, supporting the use of a single fixed resistor rather than a dynamic load.



Legend:

	Flow Rate	Peak Power
	0.17 GPM	~5 mW
	0.25 GPM	~38 mW
	0.33 GPM (Q_{eff})	**130 mW**
	0.42 GPM	~264 mW
	0.50 GPM	~329 mW
	0.67 GPM	~369 mW

Voltage flattening is visible as the sanity check described in §4.3: at $R = 20\ \Omega$, V climbs from 1.96 V at $Q = 0.42$ GPM only to 2.00 V at $Q = 0.67$ GPM despite ΔP more than doubling. Additional water energy is no longer producing proportional electrical output.

Selection

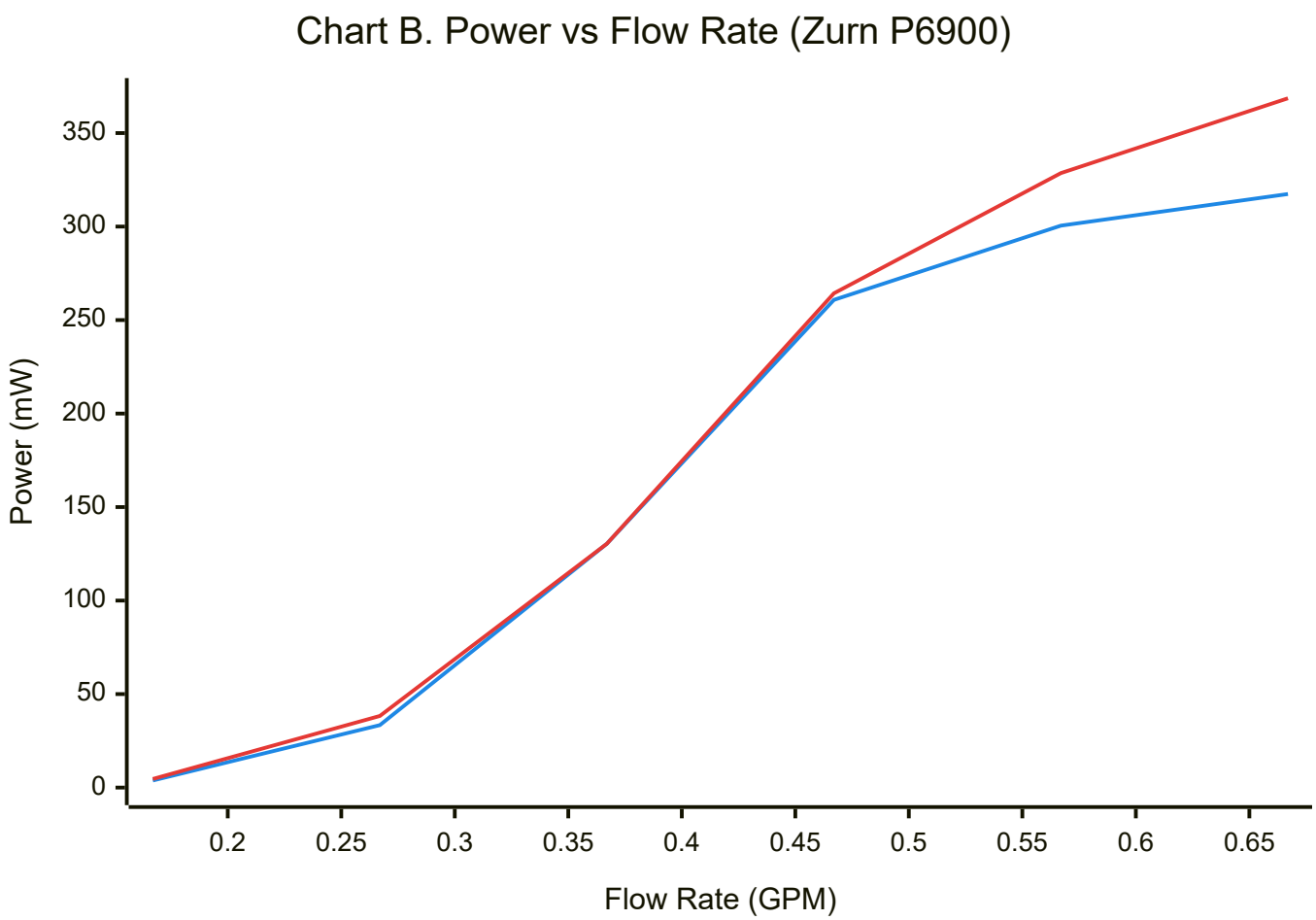
- $R_{eff} = 40\ \Omega$
- $Q_{eff} \approx 0.33$ GPM

- Peak efficiency $\eta_{max} \approx 8.0\%$
- Interior peak confirmed (R_{max} path does not apply)

At $Q = 0.67$ GPM, ΔP reaches 38.5 PSI, well above a realistic 50 PSI pressure budget once system losses are included, while power has only climbed ~22% over its Q_{eff} value. The efficiency-based Q_{eff} is the correct operating point.

Step 3: Power vs. Flow Rate (Static R_{eff} vs. Dynamic Optimal)

With R_{eff} fixed, power vs. flow rate can be plotted and compared against the theoretical ceiling of a continuously-optimized load. The two lines touch at Q_{eff} and separate as Q grows because the optimal R shifts to $60\text{-}80\ \Omega$ at higher flow rates.



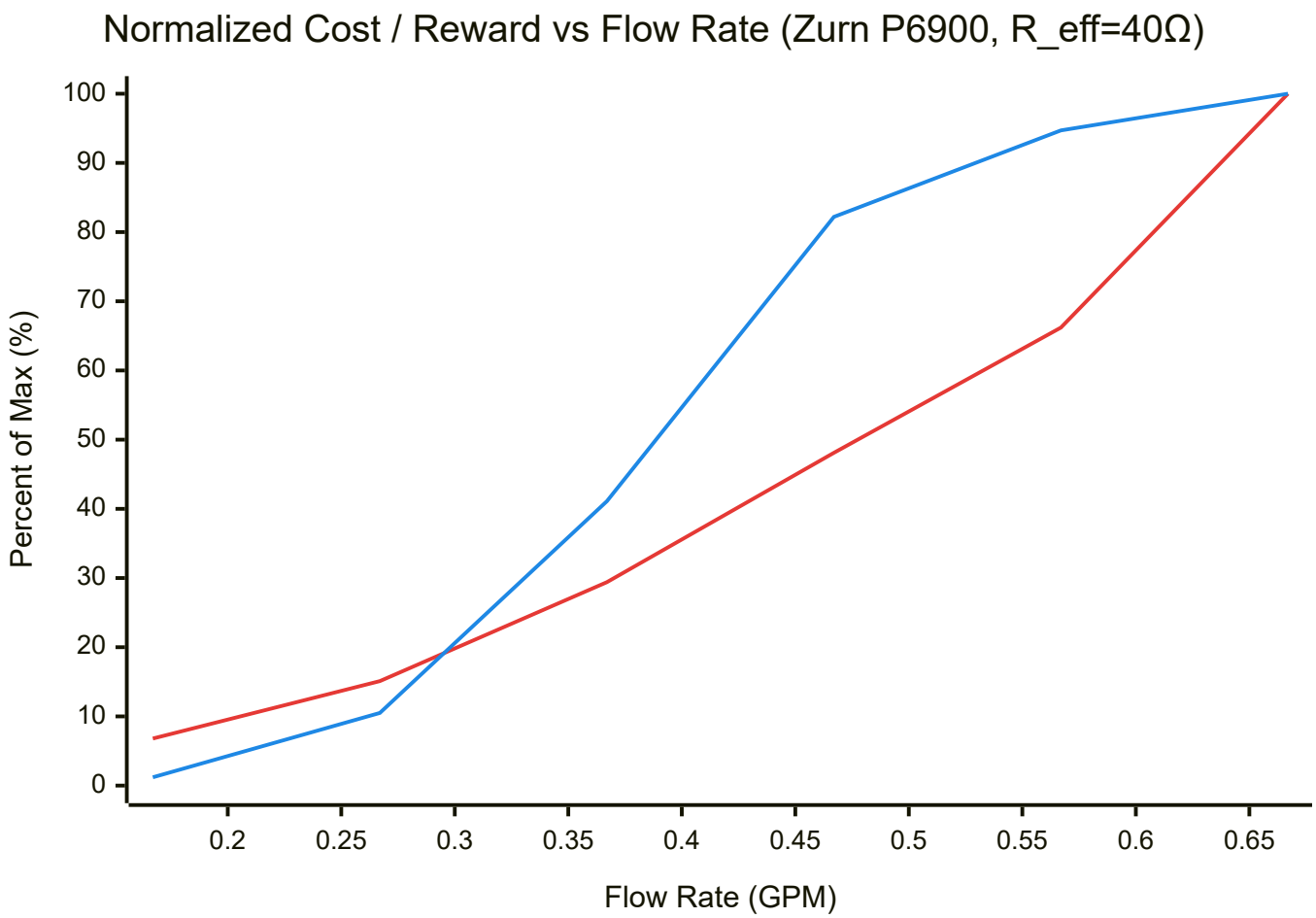
Legend:

	Load Strategy
	Static load at $R_{eff} = 40\ \Omega$
	Dynamic optimal R per Q (theoretical max)

The static-load curve captures the large majority of the theoretical maximum across the full operating range, with the largest gap (~14%) at $Q = 0.67$ GPM where R shifts furthest from $40\ \Omega$. For a fixed-load design the R_{eff} choice is robust.

Step 4: Normalized Cost / Reward vs. Flow Rate (Summary)

The standard per-turbine summary, with Cost (ΔP) and Reward (electrical power at R_{eff}) plotted on a single shared axis. Both are normalized to their respective turbine-specific maxima, so the y-axis is percent-of-max. This allows a single chart with two series instead of a dual y-axis plot.



Legend:

	Series	Conversion to Engineering Units
	Cost (ΔP as % of 38.5 PSI max)	ΔP [PSI] = cost% $\times$ 0.385
	Reward (Power at R_{eff} as % of 317.4 mW max)	P [mW] = reward% $\times$ 3.174

Interpretation: the Reward curve rises steeply through $Q_{eff} = 0.33$ GPM, reaching 41% of its maximum output while Cost is still at only 29% of its maximum. Above Q_{eff} the Reward curve

flattens while the Cost curve continues to climb, so the gap between them narrows and eventually inverts. The crossover near $Q = 0.42$ GPM (Cost = 48%, Reward = 82%) marks where additional flow begins buying proportionally less electrical return. This is the visual summary produced for every turbine in Appendix C as bench data becomes available.

4.5 Generator Output Types and Sensing Options

A turbine generator can double as a flow sensor because its output is proportional to rotor speed, which is proportional to flow rate. This subsection covers how to extract flow rate from a generator's output and which output types support it.

The table below addresses sensing from the electrical output terminals only. A separate internal-Hall option (§4.6 Method 1) works on all three output types but requires physical access to the rotor body; for an off-the-shelf regulated-DC unit this means opening the housing and moving to a custom assembly.

Output Type	Built-in Circuit	Sensing from electrical output?	How
Raw AC	None	Yes, best option	Frequency counting or power at fixed load
Unregulated DC	Diode bridge	Yes	Power proportional to flow at known load; ripple frequency also usable, though heavy filtering kills ripple amplitude
Regulated DC	Voltage regulator	No, information lost	Output held constant regardless of flow. Any flow sensing requires an internal Hall sensor (§4.6 Method 1), which means opening the housing. No longer an off-the-shelf part.

Rectification

Generator Type	Rectification
Single-coil, 2-wire	Standard bridge rectifier + smoothing cap
3-phase hub generator	3-phase full-wave rectifier (6 diodes)
Dual-coil, 4-wire	Separate rectification per coil, then combine

4.6 Sensing Methods: Priority Order

1. Hall effect sensor inside the turbine body (preferred where physically possible)

A Hall sensor mounted inside the generator housing reads the rotor's magnet directly, the same approach used in dedicated flow meters (§2). It produces a clean digital pulse train whose frequency is proportional to RPM and therefore flow rate, completely independent of the electrical load. The load can be anything (static, dynamic, open, shorted) and the Hall signal stays valid. Combined with an A1171 (~15-42 μW), this is the most power-efficient sensing method and the only viable sensing method for regulated-DC output turbines.

Caveat for regulated-DC off-the-shelf units: adding an internal Hall sensor requires opening the housing and modifying the assembly, so the product is no longer an off-the-shelf part. For sealed regulated-DC generators with a small external field leak, a clip-on external Hall detector can sometimes be used instead, but sensitivity is geometry-dependent and not reliable as a general approach.

2. Power at a known static load (secondary)

With a fixed, known load resistance, output power at a constant load scales with flow rate, but only under constant inlet pressure. The $P \propto Q^2$ relationship holds in the linear regime at fixed ΔP. In a real plumbing system, upstream pressure fluctuates with building demand, valve position elsewhere on the line, and time of day, so the same flow rate can produce noticeably different power readings. Voltage, or better yet rotor frequency via a Hall sensor, is the preferred flow proxy because it responds to rotor speed directly and is insensitive to pressure swings that move the operating point along the turbine's power curve without changing Q itself.

Power-at-fixed-load is cheapest (no additional sensor beyond the load resistor and a voltage measurement) and is acceptable where inlet pressure is known to be stable or where flow accuracy requirements are modest. Not available on regulated-DC output.

3. AC frequency or DC ripple frequency (fallback)

Raw AC zero-crossings or unregulated DC ripple frequency are both proportional to flow rate. Requires Raw AC or Unregulated DC output. Heavy downstream filtering should be avoided if using this method, as it reduces ripple amplitude. Valid fallback only if the turbine cannot physically accommodate a Hall sensor.

4.7 Open Circuit Sensing Mode

With no load connected ($R_{load} = \infty$), electromagnetic braking is removed and the turbine spins more freely, pushing the minimum detectable flow rate lower than under load. Flow rate is read from AC frequency or a Hall sensor. This mode requires a MOSFET switch to disconnect the load and a stored energy source (battery or supercapacitor) to power the sensing circuit.

The system can switch dynamically: open circuit at very low flows for maximum sensitivity; closed circuit at higher flows for power generation. See §5.3 for how this fits into the overall operating state model.

5. Hybrid Systems: Generator + Meter + Storage

5.1 Configuration Options

Option	Sensing	Power	Description
A. Generator only	Hall sensor on generator rotor	Generator output	Hall sensor reads RPM for flow; generator powers everything. Load choice is independent of sensing. Most flexible.
B. Generator power sensing	Power at fixed load	Generator output	Read power at R_{eff} as flow proxy. No separate sensor. Cheapest if a static load is acceptable.
C. Paired meter + generator + battery	Dedicated flow meter (e.g., GEMS + A1171)	Generator output + battery	Meter handles sensing; generator handles power; battery bridges zero-flow gaps. Most complete for accuracy.

5.2 Storage Strategies

Strategy	Description	Trade-off
Generator only	All power from flow	Fails at zero/very low flow
Generator + supercapacitor	MCU wakes on accumulated charge	Low cost, limited burst energy
Generator + rechargeable battery	Trickle-charge during flow	Best coverage; adds cost
Paired meter + generator + battery (Option C above)	GEMS for sensing; generator for power; battery bridges gaps	Most complete solution

> **Open question:** At what minimum flow rate can each generator reliably trickle-charge a backup battery?

5.3 Operating States & Sleep Mode

Both flow sensing and power generation are intermittent. A restroom faucet may run only minutes per day; a bottle-filling station filter may process its entire 2,000-gallon life over months. The distinction between active and sleep states dominates total energy budget.

State	Condition	Power Draw	Power Source
Active. Generating	Water flowing, load connected	MCU + Hall sensor + radio (5-50 mW typical)	Turbine output; charges battery simultaneously
Active. Sensing Only	Water flowing, load open	MCU + Hall sensor only (~0.1-1 mW)	Battery; braking minimized, extending low-flow sensitivity
Sleep	No water flow	MCU in deep sleep (~1-100 μW)	Battery only

The system can switch between generating and sensing-only states dynamically based on flow rate. Sensing-Only mode requires a MOSFET load-disconnect; the "Generator only" strategy in §5.2 (no battery) cannot support this state.

Wake detection from sleep:

- **Flow-triggered wake (preferred):** Low-power comparator monitors Hall signal or raw AC at ~1-10 μA quiescent. Detected pulse wakes MCU. Most energy-efficient.
- **Periodic polling:** MCU wakes on a timer and checks for flow. Simpler firmware but wastes energy during long no-flow periods.

Turbine selection implication: If the generator requires 0.5 GPM to produce voltage but the application needs to detect flow at 0.1 GPM, a dedicated low-power Hall sensor (e.g., GEMS + A1171, min flow 0.08 GPM) is needed alongside the generator: the paired Option C configuration.

> Specific circuit topology for sleep switching and battery management is out of scope. Design principle: size the battery to cover expected sleep duration at sleep current draw; size the turbine to replenish it within a normal active flow event.

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6. Turbine Selection Framework

6.1 Input Criteria

Input	Description	Default / Notes
Flow range (GPM)	Expected operating range; assume normal distribution	Required
Power needs	Average power when running (mW) or energy per gallon (mWh/gal)	Required

Input	Description	Default / Notes
Sensing?	Yes / No	Required
Pressure budget (PSI)	Maximum allowable ΔP across the turbine	Default 50 PSI. Legal min 25 PSI, legal max 80 PSI
Price budget	Maximum unit cost	Optional. If unknown, becomes a visualization axis

6.2 Product Tiers

Tier	What's Included	Added Requirements	Out of Scope
Raw AC	Turbine only (or regulated-DC unit with accessible coil leads)	None	Product-ready regulated DC output
Raw AC + Sensing	Turbine + Hall sensor inside body + load shutoff switch + small battery	Battery pre-charged; system detects when generation stops and switches to open-circuit sensing	Regulated DC product, specific MCU firmware

Regulated-DC units (e.g., Zurn P6900) can be used as a raw-AC tier only by opening the housing and tapping the coil leads directly. This is a benchtop / custom-housing path, not an off-the-shelf one. Making regulated DC from scratch is out of scope for turbine selection.

6.3 Selection Logic

1. **Filter by flow range:** Focus on turbines where Q_{eff} (or R_{max} operating range) overlaps the most probable flow rates.
2. **Filter by pressure budget:** Use the Cost portion of the normalized Cost/Reward chart per turbine. Eliminate turbines whose ΔP at the expected flow rate exceeds the budget.
3. **Check power output:** Confirm output meets power need at expected flow rate. Convert if needed:

$$\mathrm{mWh/gal} = \frac{P_{mW}}{Q_{GPM} \times 60}$$

> Derivation: energy per gallon = P/Q. With P in mW and Q in GPM, mWh/gal $= P\ [\mathrm{mW}] \times (1\ \mathrm{hr})/(Q\ [\mathrm{GPM}] \times 60\ \mathrm{min/hr}) = P/(60 \cdot Q)$.

4. **Apply sensing tier:** If sensing is required, confirm the turbine can physically fit a Hall sensor. Regulated-output turbines require Hall sensing and custom housing; power-based sensing is not available on those.

5. **Apply price filter.**

6.4 Two-Unknown Visualization

If two inputs are unknown, generate a scatter plot, one point per turbine, two unknown variables as axes, for visual trade-off comparison.

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7. Application Examples

Two Watts product lines are used as worked examples: **Bradley** commercial restroom faucets and **Haws** bottle-filling stations. Neither currently has IoT capability. These show how real product requirements translate into framework inputs.

7.1 Bradley: Adding IoT to a Commercial Restroom Faucet

Context: Bradley restroom faucets have no IoT capability. The goal is to add flow sensing and wireless connectivity (usage monitoring, anomaly detection, volume tracking) without wired power or battery replacement.

Input criteria:

Input	Value
Flow range	0.5-1.5 GPM (typical commercial restroom; normal distribution assumed)
Power needs	Run low-power MCU + Hall sensor during flow events; trickle-charge a backup battery
Sensing?	Yes
Pressure budget	50 PSI (default)
Price budget	TBD

Sensing required means Raw AC + Sensing tier. Specific turbine recommendation deferred until full catalogue R_{eff}/Q_{eff} data is available.

7.2 Haws: Bottle-Filling Station PID and IoT Upgrade

Context: Haws bottle-filling stations use replaceable water filters with rated gallon limits. A **Performance Indicating Device (PID)** is required under NSF/ANSI 42 (§5.3.2) and NSF/ANSI 53 (§6.3.2). PID requirements:

- Track **actual water volume** by physical displacement: a rotating turbine or positive-displacement element counting real gallons. Time-based estimation does not qualify.
- Built-in, not aftermarket.
- Operates automatically; manually resettable after filter replacement.
- Non-adjustable end-of-life setpoint.
- Clearly distinguishable visual or audible indicator when filter must be replaced (NSF does not mandate a specific color but requires a distinct warning state, described in the user manual).

Per NSF/ANSI 53, a certified PID allows filter life testing to 120% of rated capacity rather than 200% without one, enabling the manufacturer to claim a specific filter life.

The current solution is the **Digiflow 8100T-22** ($17.33, 2x AA batteries, ~3-year life replaced with the filter, LCD display).

Goals by price point:

Goal	Description
Cost reduction	Build a compliant rotary-turbine PID for less than $17.33
IoT upgrade	Add cloud connectivity or remote monitoring using flow-generated power, beyond what 2x AA can support

Input criteria:

Input	Value
Flow range	0.5-1.6 GPM (Haws filter operating range; inlet pressure TBD)
Power needs	PID only: ~3 mW active / ~45 µW sleep. IoT tier: TBD per radio selection
Sensing?	Yes, cumulative gallon count required
Pressure budget	50 PSI (default; confirm Haws inlet spec)
Price budget	<$17.33 for cost-reduction; relaxed for IoT tier

Flow range closely overlaps Bradley so the same candidate turbines apply. For cost reduction, sensing is the primary need. For IoT, the turbine must sustain a radio (BLE ~5-15 mW active) and trickle-charge a backup battery during sleep. Operating state budget follows the model in §5.3.

Appendix A: Spec Template

Category	Parameter	Value
Classification	Rotor type	Axial / Radial
	Admission	Full / Partial
	Flux type	Axial / Radial
	Coil type	Claw pole / Belt / Hub generator / Gyro
	Coil config	Single / Dual / Multi
	Output type	Raw AC / Unregulated DC / Regulated DC
	Claw poles (if applicable)	
Measured	N (turns)	
	R_{coil} (total, DC)	
	Wire gauge	
	Coil ID / OD (mm)	
Cannot Measure	B (magnetic field)	No gaussmeter
	L (inductance)	LH meter TBD
Experimental	Lumped flux coupling	From open-circuit tests
	$\eta_{turbine}$	
	c_f (viscous friction)	
	τ_0 (static friction)	
	R_{eff} (Ω)	From efficiency vs. R sweep
	Q_{eff} (GPM)	Flow rate at peak efficiency
	Min Flow = Q_{eff}?	True / False
	R_{max} (Ω)	If Min Flow = Q_{eff}; else N/A
Design Point	ΔP (rated pressure)	
	Q (rated flow)	
	P_{load} (rated power)	
	R_{load} (design load)	R_{eff} or R_{max}

Appendix B: Catalogue: Flow Meters

Meter	GEMS 238600 + A1220	GEMS 238600 + A1171	Gredia FS200A	Sea YF-S201	Uxcell	Sika VY10	Auk Mueller MT5	Digiflow 8100T-22
Picture	**take picture**	**build**						
Category	Custom Build	Custom Build	Off-the-shelf	Off-the-shelf	Off-the-shelf	Off-the-shelf	Off-the-shelf	Off-the-shelf, Display Included
Manufacturer	GEMS Setra (Plainville, CT)	GEMS Setra (Plainville, CT)	Gredia (China)	Sea (China)	Uxcell (China)	Sika (Germany)	A.u.K. Muller (Germany)	Savant / Digiflow
Turbine Type	Axial	Axial	Radial	Radial	Radial	Axial	Axial	Radial
Admission	Fully enclosed	Fully enclosed	Partial	Partial	Partial	Full	Full	Partial
Min Flow (GPM)	0.08	0.08	~0.13	~0.26	~0.1-0.15	0.16-0.18	0.05	0.21
Max Flow (GPM)	2.0 (tested to 4.0)	2.0 (tested to 4.0)	7.9	7.9	7.9	7.9	3.96	3.96
Signal Sensor	Allegro A1220	Allegro A1171	Built-in Hall	Built-in Hall	Built-in Hall	Built-in Hall (NPN open collector)	External detector clip	Built-in Hall
Output	Hall pulse (freq)	Hall pulse (freq)	Hall pulse	Hall pulse	Hall pulse	Square wave freq	Hall pulse (open drain)	LCD display
Active Power	~12 mW @ 3V	~42 µW	~75 mW @ 5V	~75 mW @ 5V	~75 mW @ 5V	45-240 mW	N/A	~3 mW
Sleep Power	N/A	~15 µW	N/A	N/A	N/A	N/A	N/A	~45 µW
Accuracy	2%	2%	±1–2%	±10%	N/A	±1% of range	±2% of range	±5%
NSF / WRAS	NSF 61, WRAS	NSF 61, WRAS	No	No	No	NSF 61, 372, WRAS, KTW, ACS	NSF approved	NSF 61/9
Max Temp	160°F	160°F	248°F	N/A	N/A	194°F (brass) / 158°F (plastic)	194°F media	170°F
Max Pressure	150 PSI	150 PSI	174 PSI	254 PSI	N/A	232 PSI (brass) / 145 PSI (plastic)	232 PSI	112 PSI
Unit Cost	~$12	~$12	$7.99	$3.92	$4.29	$9.32 (5K vol)	$16.50-$17.60	$17.33
Materials	ABS cage, Nylon 6 turbine, PEEK bearings	ABS cage, Nylon 6 turbine, PEEK bearings	ABS	Nylon	Plastic	Brass CW617N or Noryl 30% GF; SS shaft; hard ferrite magnet	PPSU body, SS shaft, ruby bearings, EPDM seals	PA66+50% GF, 304 SS axle, EPDM, ABS display
Fitting	Custom 3/8" FNPT	Custom 3/8" FNPT	G1/2"	G1/2"	G1/2"	G1/2", G3/4", 1/2" NPT, QuickFasten	Cable (3x0.25mm^2 PVC)	3/8" BSPP male
Notes	Proven in Moen U & Delta Essa. Best low-flow sensitivity.	~800x less power. Best for battery/harvesting.	Tested in-house.	Cheapest. Poor accuracy. Tested in-house.	Currently unavailable. Tested in-house.	Sapphire/PEEK bearings. Integrated flow straightener. 5kΩ pull-up needed. Tested in-house.	Two-part: turbine + detector clip.	Horizontal only. 2xAA. Sleep 15 µA.

Appendix C: Catalogue: Generators

Output Type Key

Output Type	Description
Raw AC	Unprocessed alternating current from the coil. No built-in circuitry. Frequency and amplitude both usable for sensing (see §4.5).
Unregulated DC	Built-in diode bridge rectifies AC; voltage still varies with flow. Power at a fixed load usable for sensing.
Regulated DC	Built-in voltage regulator holds output constant. Flow and pressure information lost at the output terminals. Power delivery only (or flow sense

Output Type	Description
	with internal-Hall sensor via custom housing).

Generator Comparison

Meter	Sea DB-3682	Sea DB-168 AC20V	Sea DB-168 DC5V	M6 Connector 5V	Zurn P6900-GEN	Sloan EFP40A	Toto EcoPower Insert (THP3275)	Toto TH559EDV462 (0.5 GPM Dynamo)	Toto TH559EDV561 (1.0 GPM Dynamo)
Picture							take picture		
Category	Generator	Generator	Generator	Generator	Generator (+ battery)	Full Assembly	Generator	Generator	Generator
Manufacturer	Sea / Dijiang (Foshan, China)	Sea / Dijiang (Foshan, China)	Sea / Dijiang (Foshan, China)	Generic (China)	Zurn	Sloan	Toto	Toto	Toto
Output Type	Raw AC	Raw AC	Regulated DC	Raw AC (tested via raw coil tap)	Regulated DC (Raw AC tested via raw coil tap)	Regulated DC	Raw AC	Raw AC	Raw AC
Turbine Type	Radial	Radial	Radial	Axial	Radial	Radial	Radial	Radial	Radial
Admission	Full	Partial	Partial	Full	Full	Full	Full	Full	Full
Flux Type	Axial	Radial	Radial	Radial	Axial	Axial	Axial	Axial	Axial
Min Flow (GPM)	~0.53	~0.53	~0.53	~0.6 (tested)	~0.17 (tested; 0.5 spec)	~0.5	N/A (not tested)	~0.35-0.5	~0.35-0.5
Max Flow (GPM)	~3.96	~2.64	~2.64	~3.5 (tested)	~1.5	~2.2	N/A (not tested)	~0.5	~1.0
Output	Raw AC 0-20V	Raw AC 0-20V	5V DC	5V DC	Internal battery + storage (untested)	Internal + battery backup (untested)	AC, 2-wire	AC, 9-35V	AC, 4-wire (dual coil)
Power Production	Up to 3W	Up to ~1W	Up to 0.75W	~490 mW @ Q = 3.5 GPM (tested)	~369 mW @ Q = 0.67 GPM (tested)	~mW range	~0.5W (est.)	~0.5W	~0.5W
Coil Type	Suspected hub generator, teardown required	Hub generator	Hub generator	Gyro (ring-in-ring)	Claw pole	Claw pole	Belt (wrapped)	Belt (wrapped)	Belt (wrapped)
Coil Config	TBD	Multi (3-phase)	Multi (3-phase)	—	Single	Single	Single (2-wire)	Single	Dual
R_eff (Ω)	TBD	TBD	TBD	N/A (R_max = 275)	**40**	TBD	TBD	TBD	TBD
Q_eff (GPM)	TBD	TBD	TBD	N/A (min flow = Q_eff)	**0.33**	TBD	TBD	TBD	TBD
Min Starting Pressure	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Fitting	G1/2"	G1/2"	G1/2"	G1/2" + M6	1/2"-14 NPSM	Proprietary (undersink box)	Non-separable (in faucet spout)	Proprietary (undersink)	Proprietary (undersink)
Max Temp	140°F	176°F	176°F	N/A	N/A	N/A	N/A	N/A	N/A
Max Pressure	254 PSI	174 PSI	174 PSI	N/A	80 PSI (rec.)	125 PSI	N/A	108 PSI	108 PSI
NSF	No	No	No	No	Yes	Yes	Yes (ASME, CSA)	Yes (ASME, CSA)	Yes (ASME, CSA)
Unit Cost	$13	$3.50 ($4 w/ NTC)	$3.50 ($4 w/ NTC)	$8	~$50-100+	Not sold seperately	Not sold separately	~$75+	~$75+
Tested?	No	No	No	**Yes (retest)**	**Yes (retest)**	Partial	Partial	Partial	No

Testing summary: Full bench characterization (R-sweep $\times$ Q-sweep) is complete for the **Zurn P6900** (R_{eff} / Q_{eff} demo, §4.4) and the **M6 Axial Propeller** (R_{max} demo, §4.3). All other turbines have partial test data and are pending full bench time.